Introduction to Computing

[Brian]

[Henry]

[Wayne]

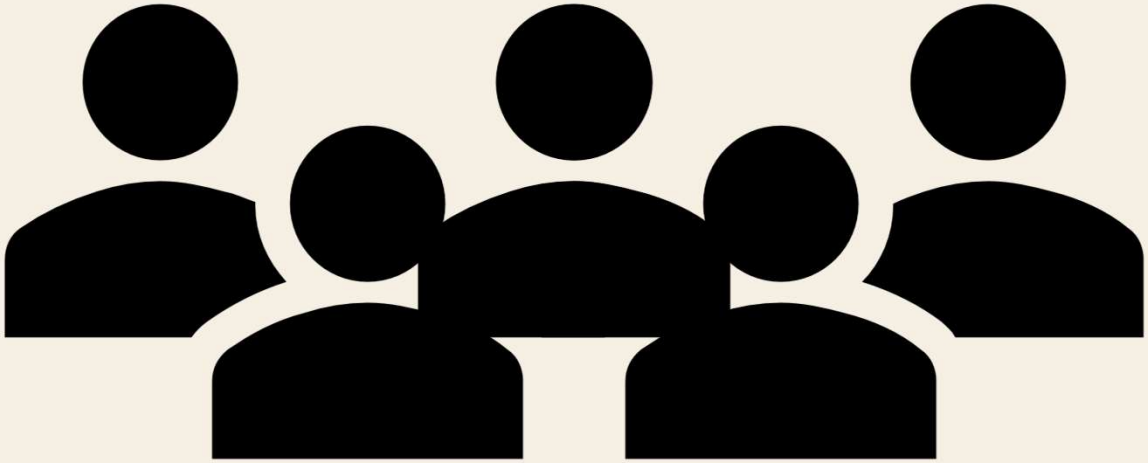
[Haneef]

[Pandeli]



Four mini demos + Meet the team

Introduction:
MP1 Computing: Wayne + Haneef
MP2 Software Development: Henry
MP3 Networking: Pandeli
MP4 Cyber-Security: Brian
Reflection and GA:I Team Task
Conclusion: Team Task



MP1: Computing

The problem : The clients were northwest architecture LTD and they needed new PC'S for 4 different roles but they need to be built specifically to match the roles and different use each staff will have for each PC. Each PC needs to meet the employee's requirements and provide suitable performance whilst still being within the budget of the company.

PC architecture :

Administrative Assistant- Henry and Wayne

We focused on building affordable and reliable computers that would still be more than capable of handling your everyday tasks such as, emails, microsoft word, excel, and just managing documents in general. This role didn’t really require powerful processors or expensive graphic cards which helped to keep costs down.

SQL database : click here

MP1: Computing p2

CAD Architect / Designer- Brian

These builds tend to need more powerful processors, reliable RAM and suitable graphics cards to handle the software, this would allow employees to be able to create detailed drawings without experiencing any slowness.

Project Manager- Pandeli

Now these computers need to be able to handle multitasking, project planning and communication, on this we focused on reliable processors, enough RAM and also fast storage so that project managers would be able to use spreadsheets with no hassle, planning software and other apps at the same time.

3D Visualisation Specialist- Haneef

These builds require the highest performance graphically because employees would potentially be working with difficult 3D models, rendering and visualising software. Powerful graphics cards, processors and enough RAM was important to be able to ensure demanding tasks can be completed to a high standard.

One Decision

For the administrative build we had personally decided to make It cost effective rather than proper high performative alternatives as it only has simple tasks to complete in comparison to the other roles and also mainly because this only needed to be able to do office basic roles.

The final build for this was £889.64 which was way under budget while still providing more than enough performance for the role

Mini demo: query live data

Open SQL database here

Run One SELECT query, short enough to explain

Show Real, correct rows. Point at one result.

```
SELECT * FROM workstationBuilds;

-- Query 8
SELECT
    Buildname.BuildName,
    SUM(Builditems.Cost) AS TotalCost
FROM workstationBuilds Buildname
JOIN BuildItems Builditems ON Buildname.BuildID = Builditems.BuildID
GROUP BY Buildname.BuildID, Buildname.BuildName
ORDER BY TotalCost DESC;
```

```
+---------+--------+-----------------------------------+----------------+
| BuildID | RoleID | BuildName                         | Source         |
+---------+--------+-----------------------------------+----------------+
|       1 |      4 | High Performance Gaming Workstation | Personal Build |
|       2 |      1 | Administrative Assistant Basic Setup | Personal Build |
|       3 |      2 | Project Manager Business Rig      | Personal Build |
|       4 |      3 | CAD Designer Architect Build      | Personal Build |
+---------+--------+-----------------------------------+----------------+
+-----------------------------------+-----------+
| BuildName                         | TotalCost |
+-----------------------------------+-----------+
| High Performance Gaming Workstation |   2118.61 |
| CAD Designer Architect Build      |   1312.57 |
| Project Manager Business Rig      |   1086.43 |
| Administrative Assistant Basic Setup |    889.64 |
+-----------------------------------+-----------+
```

MP2: Software Engineering

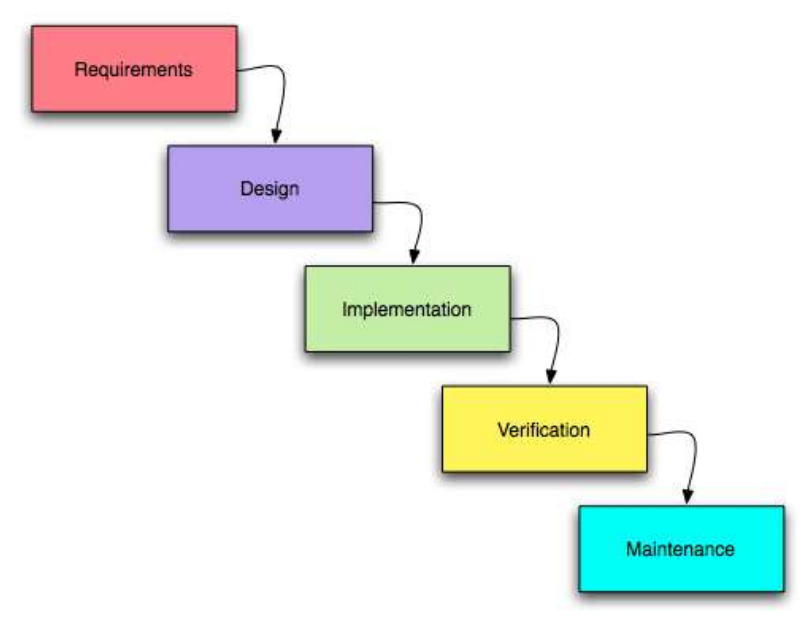
The problem: Pennine Pizza Works wants a simple computer program that can calculate customer orders more accurately. The team has been asked to design, build and test a program for them that can handle customer orders, Specifically these were their requirements for the program:

- Produce a personalized + itemized bill for the customer
- Customer details – Cost of each pizza – Total Cost – if Discount/delivery applied
- Allow choice of extra toppings and size
- Apply discount + delivery charge

How we worked

The team was split up into completing 5 individual tasks following the waterfall software development life cycle, this was how the workflow was segmented:

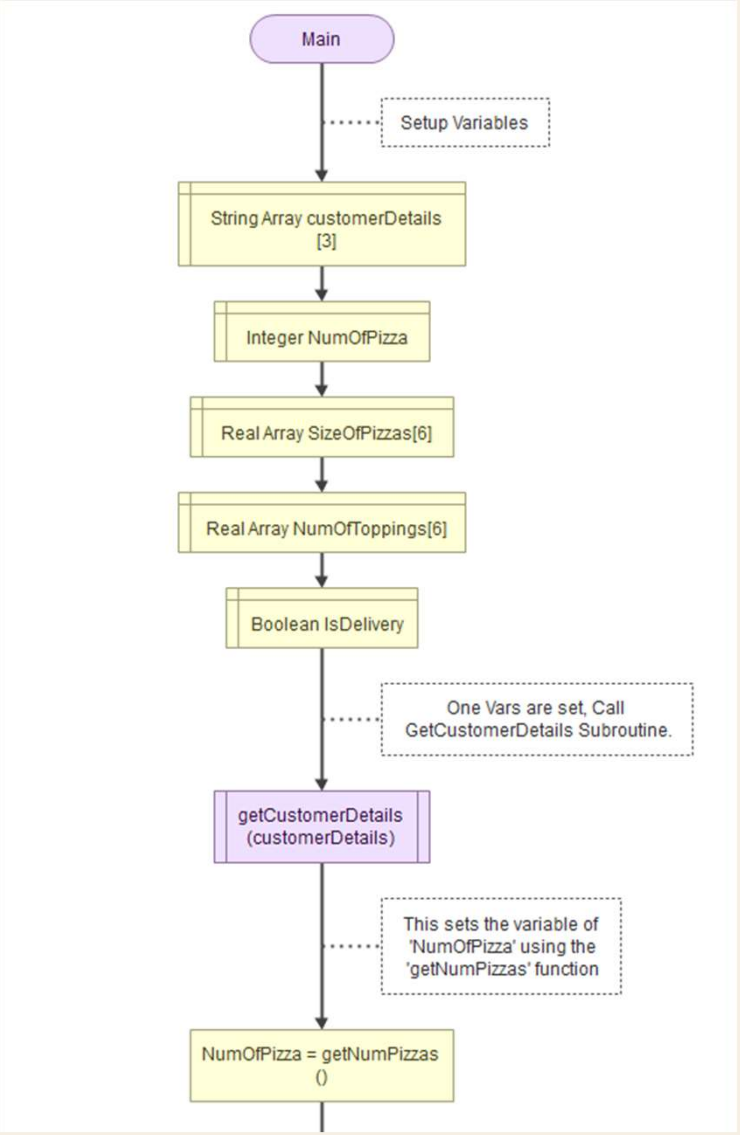
Analysis – Wayne + Pandeli
Design – Brain
Build – Henry
Testing – Haneef



Our solution: Our solution was done in Flowgorithm, and we expanded on the original task set for us in these areas where they were not otherwise specified:

- Include discount amount
- Error Handling (stringChecker)
- Store most info as arrays
- No maximum amount of toppings

Our solution was developed with good programmer practices in mind e.g. reusing functions for separate tasks.

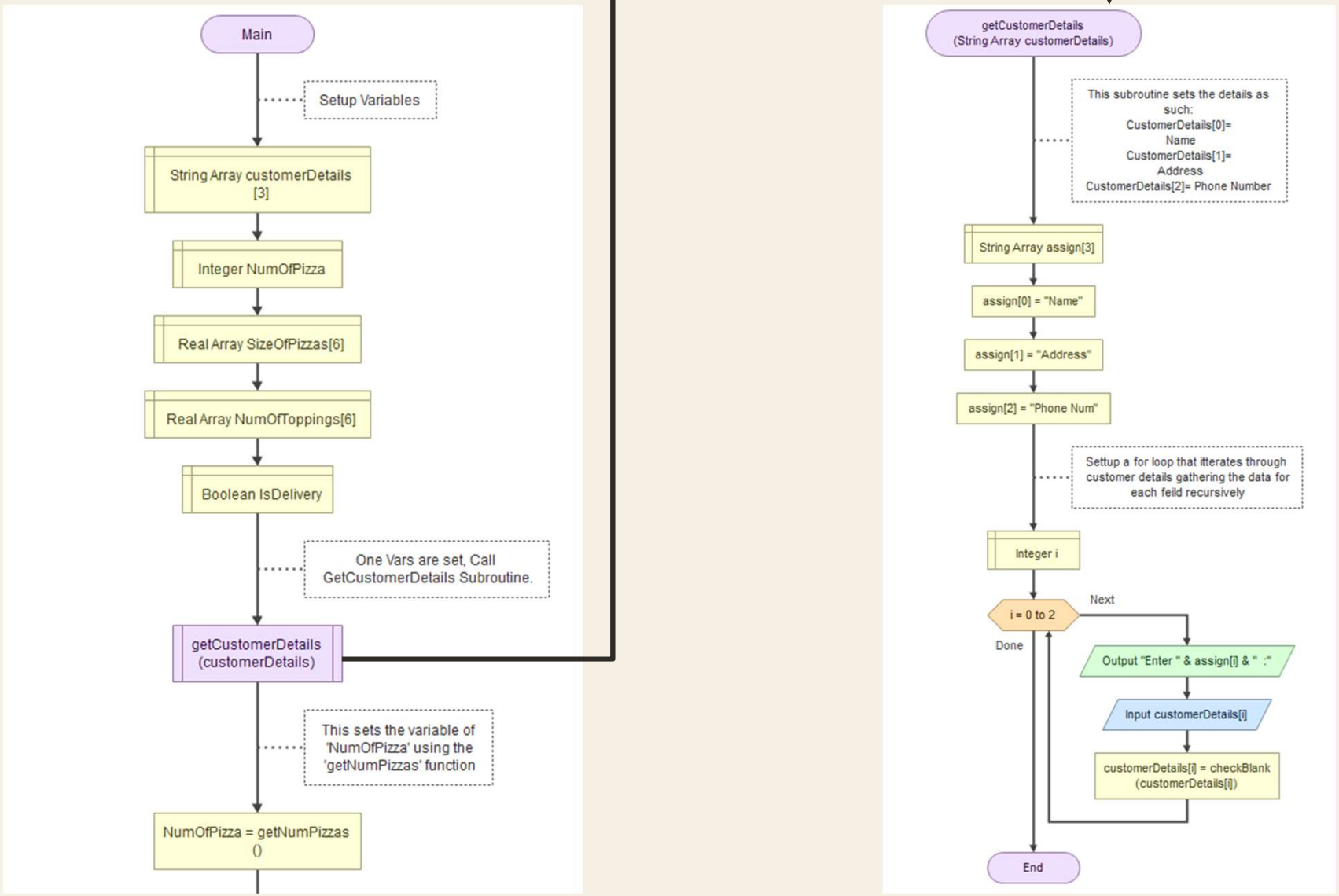


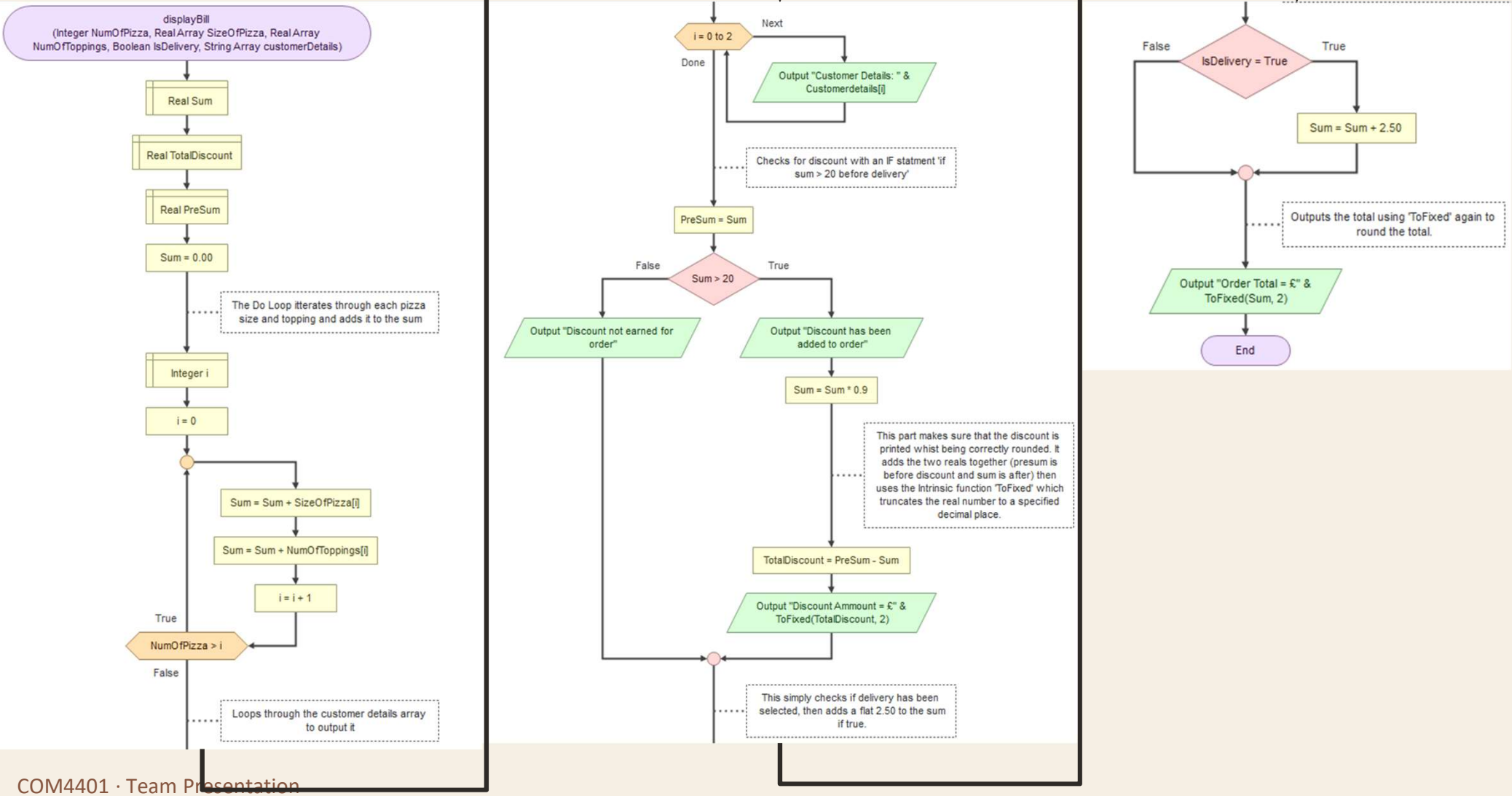
Mini demo: run the program

Open this link ☺

Run Showing typical order as test input, then show error handling

Show Output + edge case scenario


Main
Setup Variables
String Array customerDetails [3]
Integer NumOfPizza
Real Array SizeOfPizzas[6]
Real Array NumOfToppings[6]
Boolean IsDelivery
One Vars are set, Call GetCustomerDetails Subroutine.
getCustomerDetails (customerDetails)
This sets the variable of 'NumOfPizza' using the 'getNumPizzas' function
NumOfPizza = getNumPizzas ()
getCustomerDetails (String Array customerDetails)
This subroutine sets the details as such: CustomerDetails[0]= Name CustomerDetails[1]= Address CustomerDetails[2]= Phone Number
String Array assign[3]
assign[0] = "Name"
assign[1] = "Address"
assign[2] = "Phone Num"
Settup a for loop that itterates through customer details gathering the data for each feild recursively
Integer i
i = 0 to 2
Next
Done
Output "Enter " & assign[i] & " :"
Input customerDetails[i]
customerDetails[i] = checkBlank (customerDetails[i])
End

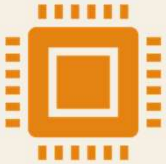

displayBill
(Integer NumOfPizza, Real Array SizeOfPizza, Real Array NumOfToppings, Boolean IsDelivery, String Array customerDetails)
Real Sum
Real TotalDiscount
Real PreSum
Sum = 0.00
The Do Loop itterates through each pizza size and topping and adds it to the sum
Integer i
i = 0
Sum = Sum + SizeOfPizza[i]
Sum = Sum + NumOfToppings[i]
i = i + 1
True
NumOfPizza > i
False
Loops through the customer details array to output it
i = 0 to 2
Next
Done
Output "Customer Details: " & Customerdetails[i]
Checks for discount with an IF statment 'if sum > 20 before delivery'
PreSum = Sum
False
Sum > 20
True
Output "Discount not earned for order"
Output "Discount has been added to order"
Sum = Sum * 0.9
This part makes sure that the discount is printed whist being correctly rounded. It adds the two reals together (presum is before discount and sum is after) then uses the Intrinsic function 'ToFixed' which truncates the real number to a specified decimal place.
TotalDiscount = PreSum - Sum
Output "Discount Ammount = £" & ToFixed(TotalDiscount, 2)
This simply checks if delivery has been selected, then adds a flat 2.50 to the sum if true.
False
IsDelivery = True
True
Sum = Sum + 2.50
Outputs the total using 'ToFixed' again to round the total.
Output "Order Total = £" & ToFixed(Sum, 2)
End

Example Results

User input is specified by highlighted yellow text

```
Enter Name  :
Genry
Enter Address  :
123 fake street
Enter Phone Num  :
0712345678
Enter number of pizzas(max 6)
2
Ok, you've selected 2 pizzas.
Select a size pizza you would like for pizza №1
1 = Small 2 = Medium 3 = Large
2
Ok for pizza 1 the size is: medium.
Select how many extra toppings for pizza no1
5
Ok you've selected 5 extra toppings
The cost for pizza 1 is: £8
Select a size pizza you would like for pizza №2
1 = Small 2 = Medium 3 = Large
3
Ok for pizza 2 the size is: large.
Select how many extra toppings for pizza no2
2
Ok you've selected 2 extra toppings
The cost for pizza 2 is: £8.5
Is this order going to be delivered? True or false only please.
true
Customer Details: Genry
Customer Details: 123 fake street
Customer Details: 0712345678
Discount not earned for order
Order Total = £19.00
```

MP3: Networking



The problem We were tasked to design and simulate a working office network for NorthWest Architecture Studio Ltd using Filius software.



Our solution Each team member created a design and have attached screenshot evidence on our report from their builds with all the testing



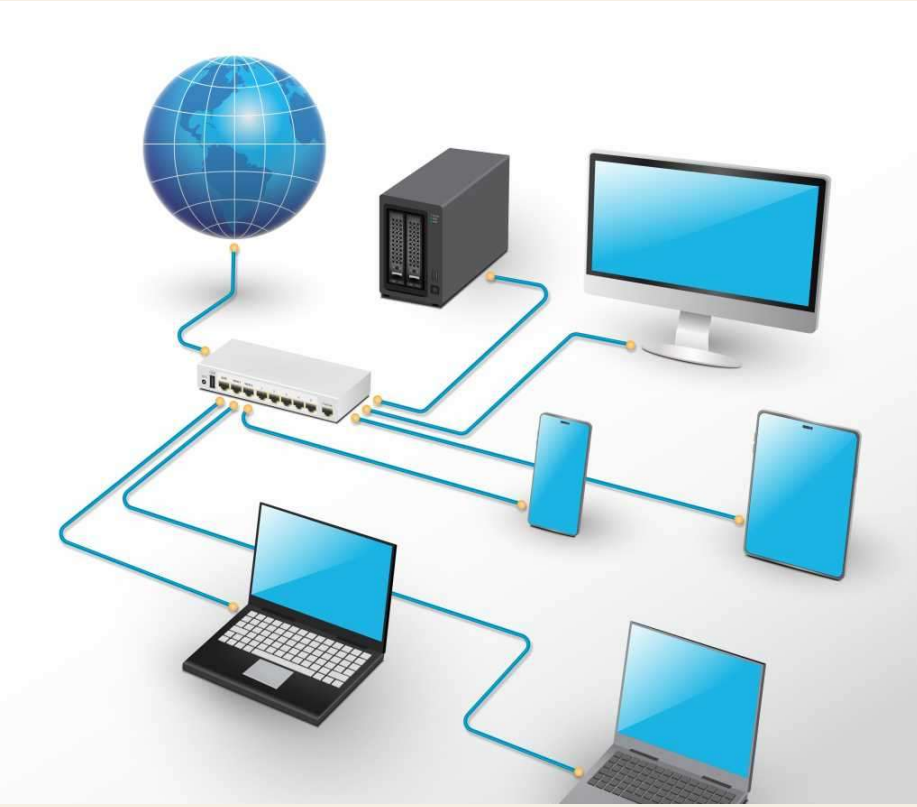
One decision We had to make a report showing the topology we were tasked to create,screenshots were attached showing successful pings between different devices in the network,test DNS, webservers and also explain at least 5 protocols and concepts from the table we were given

Mini demo: ping the network

Our filius file [here]

Run A ping or browser request: from which machine to which

Show A successful reply across the network



MP4: Cyber Security

The problem create a research paper about Cyber security

Our solution Each person research a criteria of Cyber Security create a report providing references

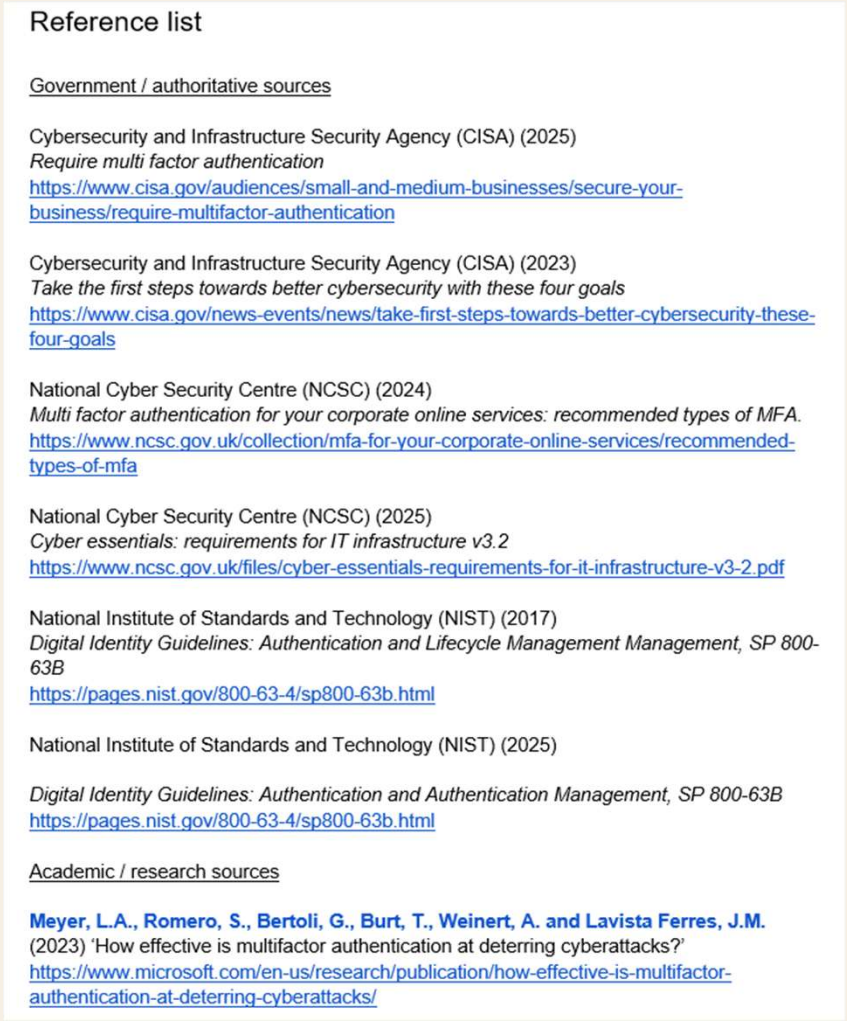
One decision Who researches which criteria of cyber security

GCS TECHNOLOGIES, INC FEATURE	Traditional SMS/Voice MFA	Microsoft Authenticator App MFA	Passwordless/Passkey Authentication
Security Level	Moderate	High	Very High
Ease of Use	Moderate	High	High
Susceptibility to Phishing	High	Low	Very Low
Susceptibility to SIM Swapping	High	None	None
Biometric Support	No	No	Yes
Hardware Requirement	Phone	Smartphone	Smartphone or Security Key
Offline Capability	Yes	Yes	Yes
User Experience	Moderate	High	High

Mini demo: MFA

Link Sample of Cyber Security document

Point I decided to use my part of the research paper since it is the part of our research paper, I'm most familiar with



Reference list

Government / authoritative sources

Cybersecurity and Infrastructure Security Agency (CISA) (2025)
Require multi factor authentication
https://www.cisa.gov/audiences/small-and-medium-businesses/secure-your-business/require-multifactor-authentication

Cybersecurity and Infrastructure Security Agency (CISA) (2023)
Take the first steps towards better cybersecurity with these four goals
https://www.cisa.gov/news-events/news/take-first-steps-towards-better-cybersecurity-these-four-goals

National Cyber Security Centre (NCSC) (2024)
Multi factor authentication for your corporate online services: recommended types of MFA.
https://www.ncsc.gov.uk/collection/mfa-for-your-corporate-online-services/recommended-types-of-mfa

National Cyber Security Centre (NCSC) (2025)
Cyber essentials: requirements for IT infrastructure v3.2
https://www.ncsc.gov.uk/files/cyber-essentials-requirements-for-it-infrastructure-v3-2.pdf

National Institute of Standards and Technology (NIST) (2017)
Digital Identity Guidelines: Authentication and Lifecycle Management Management, SP 800-63B
https://pages.nist.gov/800-63-4/sp800-63b.html

National Institute of Standards and Technology (NIST) (2025)

Digital Identity Guidelines: Authentication and Authentication Management, SP 800-63B
https://pages.nist.gov/800-63-4/sp800-63b.html

Academic / research sources

Meyer, L.A., Romero, S., Bertoli, G., Burt, T., Weinert, A. and Lavista Ferres, J.M. (2023) 'How effective is multifactor authentication at deterring cyberattacks?'
https://www.microsoft.com/en-us/research/publication/how-effective-is-multifactor-authentication-at-deterring-cyberattacks/

What links the four projects

How they connect All these mini projects connect different essential and important aspects of computing
What went well We were quick when it came to getting the work done
What was hard or unfinished I challenge we faced was communication
What we learned working as a team is essential with project work and communication is important

How we used and checked GAI

MP1 Not used→ No Ai was used or needed to be used for the builds, nor the complete database.

MP2 Not used at all → No Ai was used to create or help the creation of the Flowgorithm application nor the document.
MP3 Not used at all → No Ai was used to create the networks

MP4 Not / partly used → during our research we avoided using AI, but some members used AI when gathering their data since we saw errors in their referencing

Conclusion and questions ?

Takeaway overall, we as a team believe that we could have communicated as a team better, and motivated each other to push through with doing our work on time efficiently

Q&A does anyone have questions